

Challenge 9: Intelligent Complaint Classification Using Localized Transformer Architectures

Customer support organizations receive thousands of consumer complaints every day, making manual complaint classification slow, inconsistent, and prone to human error. Traditional complaint handling methods often struggle to accurately categorize unstructured text because they fail to capture context, intent, sentiment, spelling variations, and localized expressions. As complaint volumes continue to increase, these limitations result in delayed responses, incorrect complaint routing, increased operational costs, and reduced customer satisfaction. Existing approaches also perform poorly when handling diverse communication styles, including regional languages and informal expressions, limiting their effectiveness in real-world environments. Addressing these challenges is essential to improve complaint management, support timely decision-making, enhance customer experience, and ensure that all users receive fair and efficient service regardless of how they communicate.